Q18: Discussion Questions

DLS 2026 — Kukla

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1. Model activations predict responses in sensory cortex but much less so in high-level reasoning areas. What does this asymmetry suggest about where the brain analogy is load-bearing?

2. Backpropagation is biologically implausible, yet it works spectacularly. Does that make the "brains learn differently" objection scientifically interesting or practically irrelevant? Argue a side.

3. If an LLM matches human behavior on a reasoning benchmark but by a different mechanism, in what concrete situations would the mechanism difference actually matter?

4. Neuromorphic hardware promises brain-like energy efficiency. What would have to be true for it to beat GPUs on a real task, not just on power?

5. Which single feature of biological computation, if we could import it faithfully, would you bet most improves current AI—and why?